# Introduction to Computer Networks and the Internet
# COSC 264

## Local Area Networks and Ethernet

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# Outline


1. LANs
   - LAN Standards
   - Physical Topologies
2. MAC Fundamentals
3. Orthogonal MAC Schemes: FDMA and TDMA
4. Random Access Protocols
5. Ethernet
   - Ethernet Frame Format
   - PHY
   - Half-Duplex Ethernet MAC Protocol
6. Bridges, Switches
   - Repeaters and Hubs
   - Bridges and Switches

# Preliminaries

- The following slides are based mainly on [28], [25]
- The older, but still very good book [6] covers some related topics, book is available in pdf format here:

  `http://web.mit.edu/dimitrib/www/datanets.html`

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6 Bridges, Switches

# Local Area Networks (LANs)

- LANs are packet-switched networks
  - Packets are often called **frames** in LAN context
- They have limited geographical extension, usually $\leq$ 1 km
- Often controlled by only one owner / administrative entity
- They offer:
  - A shared transmission medium to multiple stations
  - Low cost for station attachment
  - Higher rates than usually experienced in wide-area networks
- Some application areas:
  - Connect desktop computers to share files, emails, ...
  - Allow several computers to share printers, file servers, ...
  - Interactive video or telephony between local users
  - Providing Internet access

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# LAN Protocol Architecture

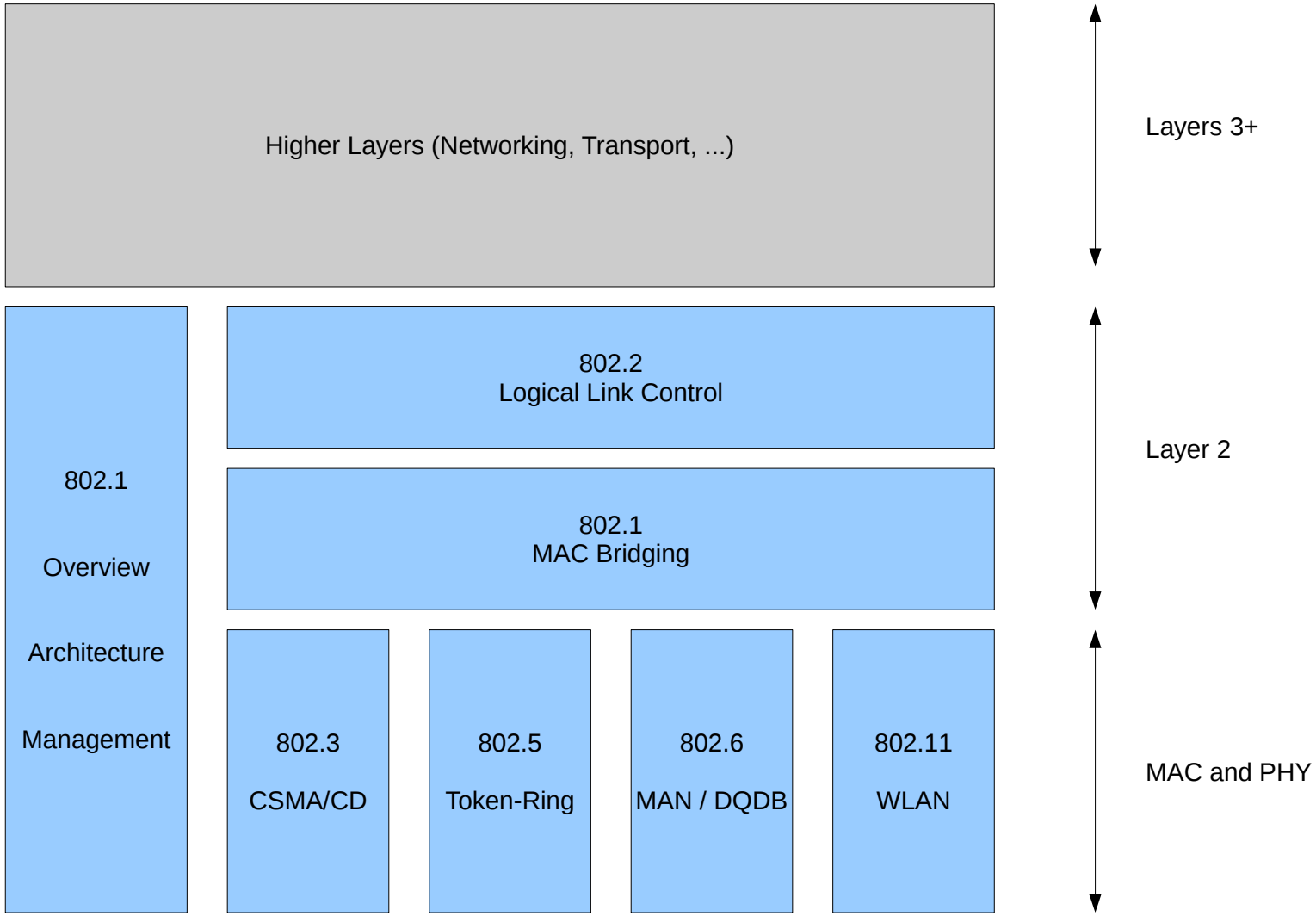
- LAN standards typically specify the following layers:
  - **Physical layer (PHY)**, specifying transmission media, network adapters, modulation schemes, data rates and network topologies
  - **Medium Access Control (MAC)** sublayer, specifying how stations share the transmission medium
  - Optional: **Link layer** (often called logical link control), providing error-control, flow-control, etc.
    - Error control: error detection, retransmissions, error-correction coding
    - Flow control: stopping a fast transmitter from overwhelming a slow receiver with more data than it can currently process

    Error-correction coding often done in hardware, integrated in PHY
- This is the protocol architecture followed by the IEEE standards, which are the dominant LAN standards

# The IEEE 802 Standards Series

- IEEE = Institute of Electrical and Electronics Engineers
- IEEE is a professional association, not a standards body
- Nonetheless, ISO adopted several IEEE standards
- Some important standards (series):
  - IEEE 802.1: Overview, Bridging, Network Management, ...
  - IEEE 802.2: Logical Link Control (LLC) [23]
  - IEEE 802.3: Ethernet [14]
  - IEEE 802.11: Wireless LAN [16]
  - IEEE 802.15: Wireless Personal Area Network (WPAN), incl. Bluetooth [15], IEEE 802.15.4 [21], ...
- There are *many* more ...
- See `http://standards.ieee.org/getieee802`

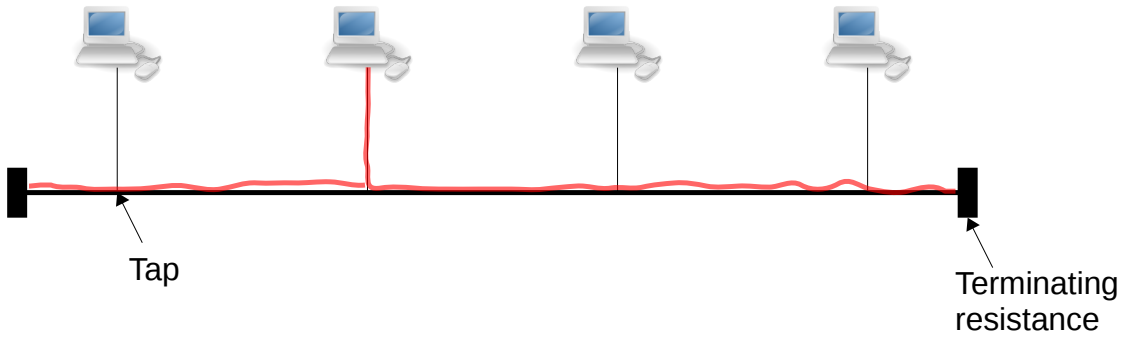
# The IEEE 802 Standards Series (2)



# Outline

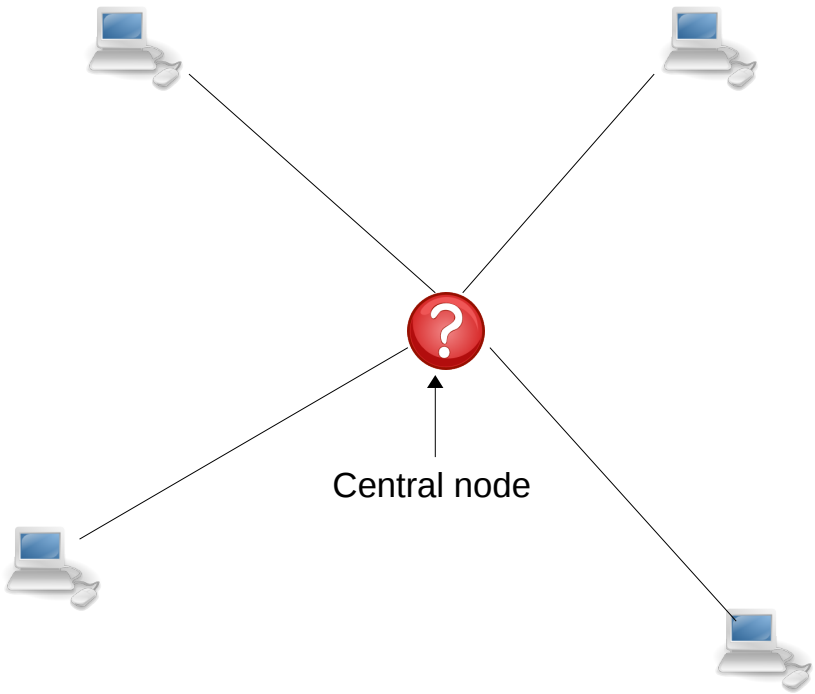
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# Bus Topology




- Stations are attached via tap line to bus
- Bus is a broadcast medium, signals are heard by all stations
- Parallel transmissions collide, all frames are garbled
- Frames include destination address field to indicate intended receiver
- Bus length usually limited to a few dozens / hundreds of m, e.g. to prevent excessive signal attenuation
- By cascading several buses (using branch points) more complex broadcast toplogogies can be built (e.g. trees)
- Examples: classical Ethernet, Token Bus (IEEE 802.4 [13])

# Star Topology




- Stations are attached to a central unit, often with separate cables for transmit and receive directions
- Central unit can act either as a repeater or as a bridge/switch
- A repeater copies incoming signals to *all* other outputs
- A bridge/switch copies incoming frame only to output where destination is
- Discussed in more depth later

# Wireless Topologies

- Wireless transmission media are much more prone to errors than wired media [26], [33], bit-error rates of $10^{-2}$ are not uncommon
- Wireless links are influenced by propagation environment
  - Doors, moving surfaces, walls, ...
- A station cannot necessarily hear **all** other stations

**Important Property**

Wireless transmission channels are time-varying, the links are volatile and unpredictable, there is usually no static wireless topology

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# Medium Access Control (MAC)

- MAC protocols are schemes which allow a number of users to coordinate the access to a common channel
- MAC protocols are a vast subject, some references (not even the tip of the iceberg!) are: [12], [27], [22], [8], [34], [29], [2], [18], [20], [9], [30], [17], [31], [32] [4], [5], [3], [10]
- The MAC layer is often regarded as a separate sub-layer between PHY and link-layer
- This view is supported by the fact that the MAC has a distinguished task not covered by any other layer
- MAC protocols are heavily influenced by the properties of the underlying transmission medium

# MAC Definition

- We are given:
  - A number of users / stations wishing to communicate
  - A shared communications channel / resource that can only be used by one station at a time
  - No other means for information exchange between stations

**Definition**

MAC protocols are rules by which distributed stations coordinate access to a common channel to share it efficiently and in a manner satisfying given performance requirements

Example: 100 blind persons in a room – how to distribute right to talk?

# Important Assumptions

- The shared channel is a broadcast medium, i.e. transmission of one station is heard by *all* other stations
  - Not necessarily true for wireless transmission media
- In case of parallel transmissions all contending transmissions are garbled, i.e. cannot be reliably decoded
  - Not necessarily true, e.g. for wireless transmission media

# MAC Design Desiderata

- Small *medium access delay*: time between arrival of frame to empty station and start of successful transmission
  - Depends on overheads: collisions, waiting times, ...
  - For lightly loaded medium a small access delay is desirable
- Fairness and fair re-use of unused resources
- Efficiency: low overhead, high throughput
- Stability: increasing load should not decrease throughput

Note: you usually do not get all of these at the same time ...

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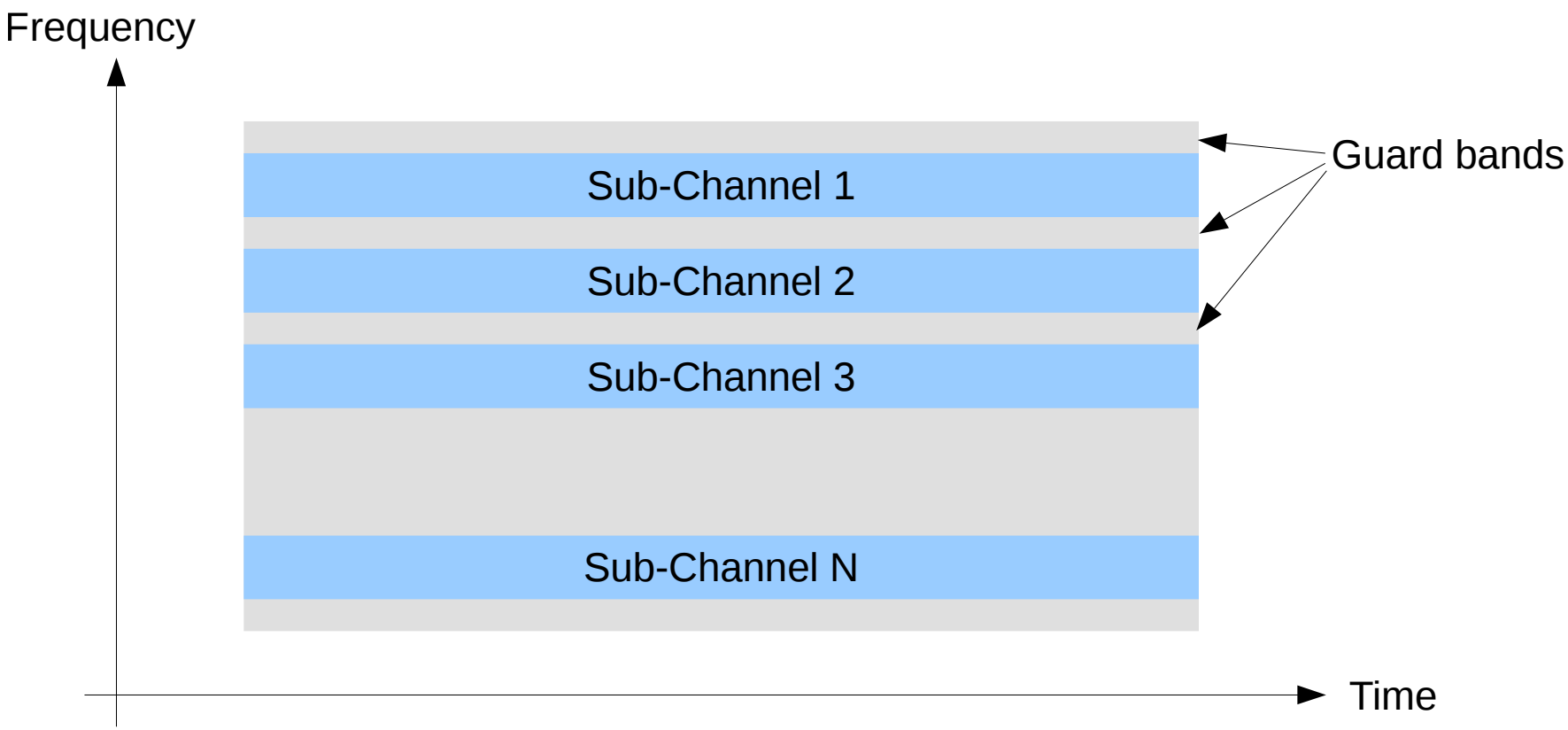
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# Orthogonal Schemes

- In *orthogonal schemes* the behavior of one station does not influence the behavior / throughput / transmission success / . . . of other stations
- Orthogonal schemes allow for collision-free transmission
- The four main (mostly) orthogonal schemes are:
  - FDMA = Frequency Division Multiple Access
  - TDMA = Time Division Multiple Access
  - SDMA = Space Division Multiple Access
  - CDMA = Code Division Multiple Access
- We will not discuss SDMA and CDMA

# Frequency Division Multiple Access (FDMA)



# FDMA (2)

- The given channel bandwidth is subdivided into *N sub-channels*
- Between the sub-channels and at the fringe of the channel there are *guard bands*:
  - Reduction of adjacent-channel interference, robustness against imperfect frequency synchronization
- A sub-channel is *exclusively* assigned to a station *i* on a longer-term basis for transmission of data, **no other station is allowed to transmit on this channel**
- To receive data, a station must:
  - Either possess a separate receiver for each channel, or
  - Have a single tunable receiver that must be switched to a specific channel before data can be received on it
    - Problems: coordination/rendez-vous, tuning times

# FDMA (3)

- If totally available bandwidth is $B$ $\left[\frac{b}{s}\right]$, station $i$ is assigned $\frac{1}{N}$ of $B$ on a long-term basis (neglecting guard bands)
- Medium access delay for a new packet arriving to an empty station $i$ is always zero, since $i$ can start transmission immediately without risk of collision
- If a packet has size $\frac{B}{N}$ bits, its transmission takes one second:

$$E\left[\text{Transmission Delay}\right] = 1$$

- Its total completion time is the sum of transmission and access delay, which here is equal to the transmission delay

# FDMA – Advantages

- $N$ stations can transmit in parallel
- No need for time synchronization between the $N$ transmitters

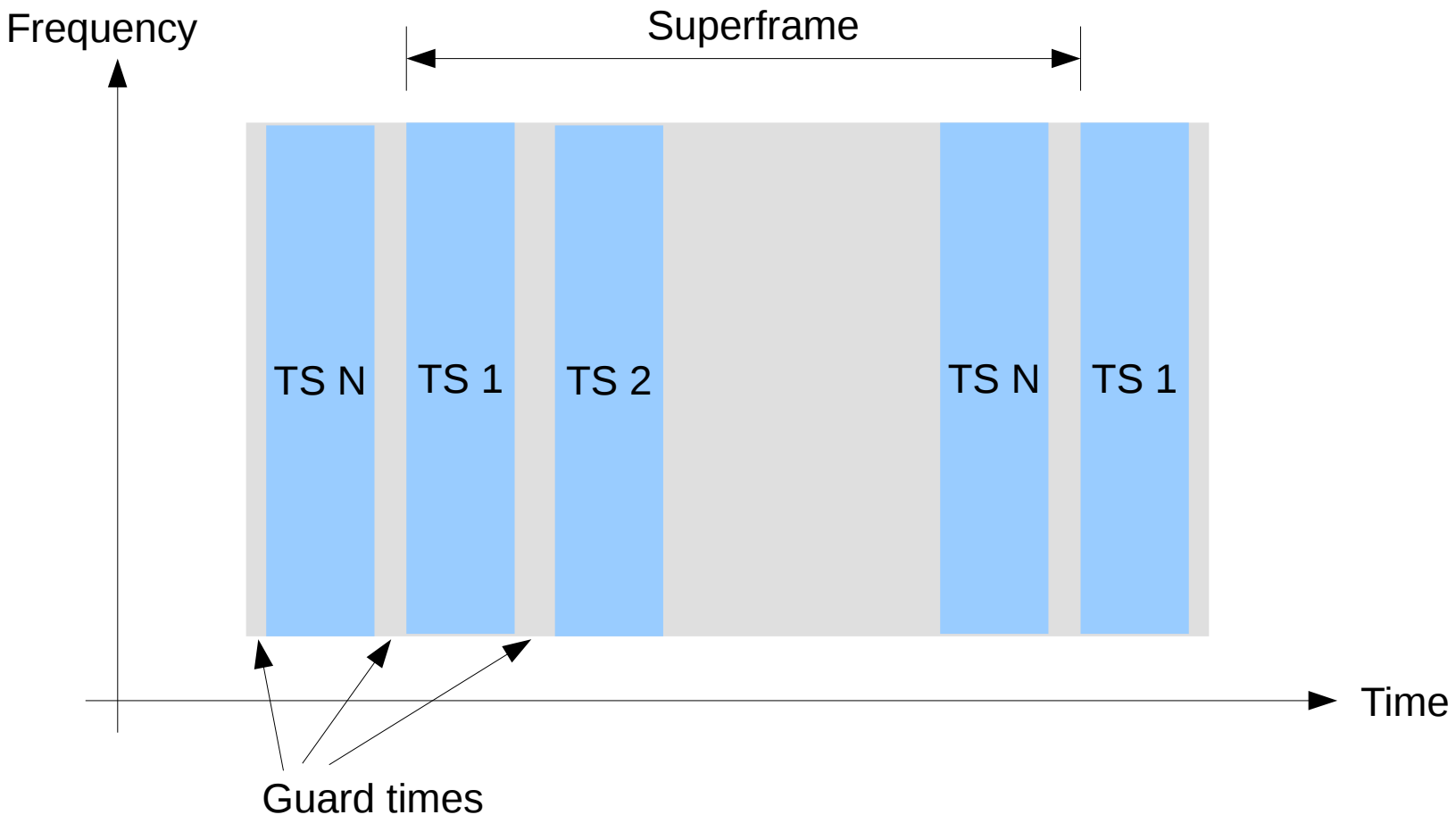
# FDMA – Disadvantages

- Need for $N$ receivers or tunable receivers increases complexity
- Frequency synchronization required
- No re-use: channel not used by owner can't be used by others
- Coordination and shared state required for allocating subchannels

**Conclusion**

FDMA is good for CBR but bad for VBR traffic

# Time Division Multiple Access (TDMA)



# TDMA (2)

- Each station uses the whole frequency band (except some guard bands at the fringe of the spectrum), but only at certain times:
  - Time is subdivided into *superframes* of duration $T_{SF}$
  - Each superframe is subdivided into $N$ *time-slots*
  - There are short *guard times* between time slots
  - One or more time slots are assigned exclusively and on a longer-term basis to a station $i$ for transmission
- Stations must be time-synchronized to avoid overlapping transmissions, guard times are required to compensate (small) synchronization errors

# Completion Time in TDMA

- Neglecting guard times, each station having a slot gets the full channel bandwidth $B\ \left[\frac{b}{s}\right]$ for a fraction of $\frac{1}{N}$ of time
- Assume that:
  - station $i$ owns one time slot
  - $T_{SF} = 1$ second
  - a time-slot suffices to transmit $\frac{B}{N}$ bits (we ignore guard times)
  - a packet of $\frac{B}{N}$ bits arrives at random time to empty station $i$
- Medium access delay = waiting time until station $i$'s next slot starts

$$E\left[\text{Access Delay}\right] = \frac{T_{SF}}{2} = 0.5\ \ [s]$$

# Completion Time in TDMA (2)

- The time to transmit the packet (transmission delay) is $\frac{1}{N}$ seconds
- Assuming no channel errors we have:

$$\begin{aligned} E\,[\text{Completion Time}] &= E\,[\text{Access Delay}] + E\,[\text{Transm. Delay}] \\ &= 0.5\ [s] + \frac{1}{N}\ [s] \leq 1\ [s] \end{aligned}$$

- For $N > 2$ this is a true inequality!

**Conclusion**

With TDMA we can start later and finish sooner than with FDMA!!

# TDMA – Advantages and Disadvantages

- Advantages:
  - It is easier to achieve asymmetric bandwidth assignments in TDMA than in FDMA: using multiple time-slots is much simpler than transmitting on multiple frequencies in parallel
  - TDMA tends to have better completion times than FDMA
  - No tunable receivers needed
- Disadvantages:
  - Tight time-synchronization between stations required
  - High expected access delay even in otherwise idle systems
  - Not possible to re-use unused time slots
  - Coordination and shared state required for allocating time slots

**Conclusion**

TDMA is good for CBR but bad for VBR traffic

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# Random Access Protocols

- Random access protocols:
  - Do not attempt to reserve channel resources for longer time
  - Do not require a central station or (much) shared state
  - Do not access the medium at predictable times
  - Often have low complexity (e.g. only little coordination, if any)
  - Typically involve some random element
- Random access protocols are used standalone and also as building blocks for more complex MAC protocols, e.g.:
  - Mobile broadband technologies use random access protocols for stations to request call setup

**Important Point**

Random access protocols accept risk of collisions to save coordination overhead and have overall improved efficiency! How well this trade-off works depends for example on the offered traffic load.

# ALOHA

- ALOHA [1] is one of the earliest MAC protocols, developed $\approx$ 1970 at the University of Hawaii
- Assumptions:
  - $N$ uncoordinated transmitters
  - One receiver (e.g. a base station)
  - If two packets overlap at receiver, a *collision* occurs

# The Pure ALOHA Protocol

- When a new frame arrives at an idle station:
  - A checksum is computed and appended to the frame
  - The frame is then **transmitted immediately**, there is no coordination with other stations
  - An *acknowledgement timer* is started
- The receiver sends an *immediate ack* upon successful reception of a frame – upon collisions or transmission errors it remains quiet
- If the transmitter receives an ack before the ack timeout, the frame is removed and the ack timer is canceled

# The Pure ALOHA Protocol (2)

- When ack timer expires, transmitter enters **backoff mode**:
  - The transmitter chooses a random backoff time
  - It waits for this time without further action
  - At backoff timer expiry the frame is re-transmitted
  - The ack timer is set again, backoff mode is left
  - **Question**: why is the backoff time chosen randomly?
- When number of failed trials exceeds threshold, frame is dropped
- The precise choice of random distribution for backoff times is critical for delay, throughput and stability! [11], [19]
  - Often the random distribution depends on the number of subsequent collisions seen by the frame
  - An example backoff strategy will be discussed later!
- When a new frame arrives to backlogged station, it is stored in queue and served after all previous frames
- When transmission of a frame is completed and queue is not empty, then the next frame is transmitted after random backoff

# Advantages of Pure ALOHA

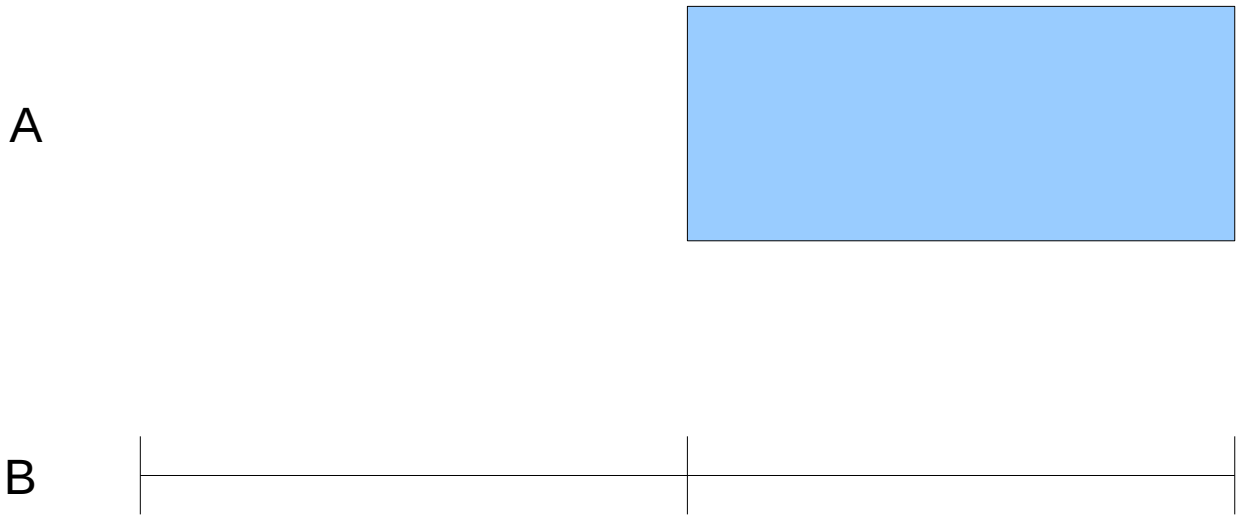
- Quite simple to implement
- If network load is small:
  - New frames are sent immediately $\Longrightarrow$ access delay is zero
  - They can use the full channel bandwidth
  - And the probability of collision at the receiver is low

**Conclusion**

For low network loads most packets can have the minimum possible completion time

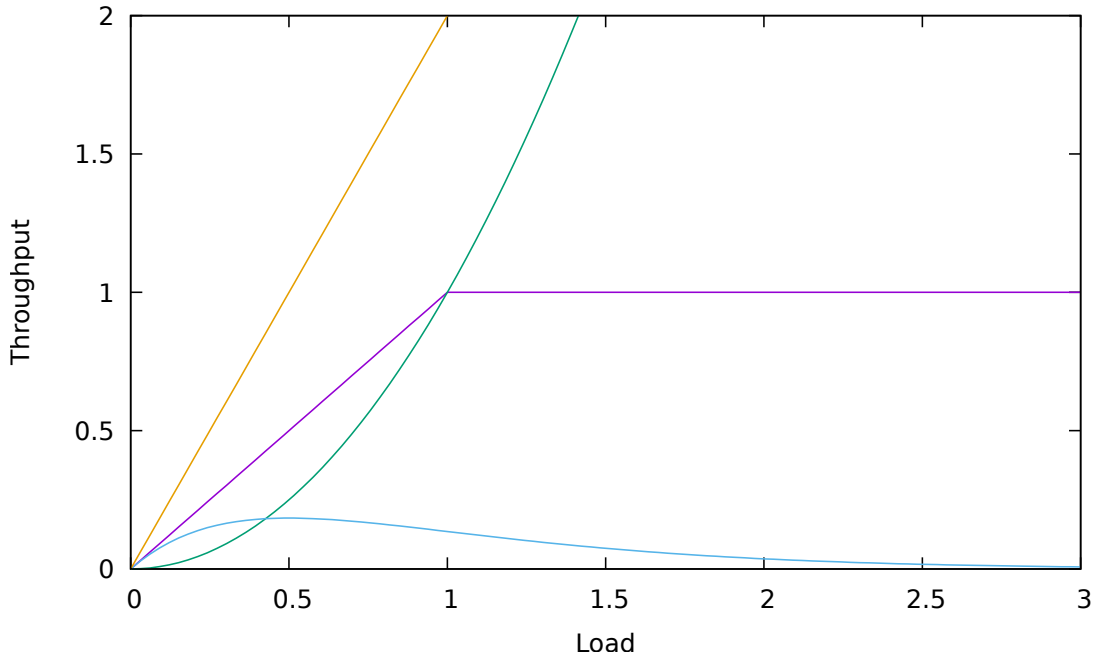
# Disadvantages of Pure ALOHA

- Consider stations *A* and *B* sending frames of same length:




- When *B* starts its frame during the (two frame times long) *vulnerability period* of *A*'s frame, the frames collide
- When more stations are active the collision probability increases

# Throughput




- Consider a system with many transmitters with equal average load, each generating packets at random times
- Aggregate load is given, all packets have the same length, packet transmission time is $\tau$
- Define load as the aggregate (average) number of packets generated during time $\tau$, througput as the (average) number of successfully received packets during time $\tau$
- **Q:** Which of these (idealized) curves is the throughput of ALOHA? And which one the throughput of FDMA? What about the others?

# The CSMA Family of Protocols

- CSMA = Carrier Sense Multiple Access [17], [31]
- Common assumption: all stations can determine the state of the medium (almost) instantaneously:
  - **Busy**: at least one station is currently transmitting
  - **Idle**: no station is transmitting

  This operation is called **carrier-sensing** (CS) or **clear-channel assessment** (CCA)
- Common approach: **Listen-before-talk** (LBT)
  - Before station transmits a frame, it performs CS operation
  - If channel is busy, station defers transmission according to one of several possible strategies
  - The maximum number of deferrals (or **backoffs**) a station might experience for a frame is often bounded
- CSMA doesn't eliminate collisions, but reduces their rate or impact

# The CSMA Family of Protocols (2)

- CSMA approach is especially useful when medium propagation time is small compared to frame length, since other stations notice transmission quickly after it started
  - Typically satisfied in LANs, propagation delay is small
  - Collisions can only occur when two stations start transmitting at almost same time (time difference smaller than propagation delay)
- When propagation time is large compared to frame length, the sender might have already stopped transmission when receiver senses a busy carrier for the first time
  - Example: multi-access satellite configurations
  - Here LBT is almost useless, ALOHA is reasonable

# Nonpersistent CSMA

- After getting a new frame for transmission, a station first performs a carrier-sense operation
- If a station senses a busy medium, it:
  - Draws a backoff time from a given random distribution
  - Defers from channel activities during backoff time, and
  - After the backoff time it senses the medium again and starts over
- If station detects idle medium, it starts transmitting immediately
- In case of collision a backoff time is chosen, process starts over
  - **Question**: How to diagnose collisions?
- Performance problem: with high probability the medium is idle for some time after a transmission has finished, lowering utilization

# $p$-persistent CSMA

- Be $p \in (0, 1)$ a parameter known to all stations
- If a station senses a busy medium, it defers until the end of the ongoing transmission, when medium becomes idle
- A station divides time on idle medium into small time slots
  - At the beginning of a time slot a station performs a random experiment: with probability $p$ it starts transmission, with probability $1 - p$ it defers for one further slot
  - When station defers, it checks medium during remaining slot time: when another station started transmission, station waits for end of this transmission and starts over
  - Time slot just large enough to accommodate these activities
- In case of collision, process starts over
- **Question**: How would you choose $p$?
- Performance problem: again, medium will be idle for some time after previous transmission has finished

# 1-persistent CSMA or CSMA/CD

- CSMA/CD was chosen for classical Ethernet
- If station senses busy medium, it defers until end of transmission
- When medium is / becomes idle, station sends unconditionally
  - This avoids idle times after previous transmission
  - But if two or more stations start, we surely have collision and we need **collision resolution procedure**
- While transmitting, sender tests channel for collisions
- In case of a collision:
  - Transmission is aborted
  - A jamming signal is sent to inform all stations about collision
  - A collision resolution procedure is started, e.g.:
    - Backoff schemes (used in Ethernet and WLANs)
    - Tree algorithms [7]

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# Ethernet

- Ethernet is a packet-switched LAN technology, invented in 1973 by Metcalfe and Boggs at Xerox PARC
- Ethernet became the dominant wired LAN technology, early competitors like Token-Ring, Token-Bus vanished
- Standardized by IEEE ([14] and various amendments)
- Its name refers to the “Ether” that was believed to be the necessary medium for electromagnetic wave propagation
- It offers:
  - High data rates (10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, 100 Gbps)
  - Cheap and mature components
  - Large market penetration, many different vendors
- These slides are mainly based on [28, Chap. 16]

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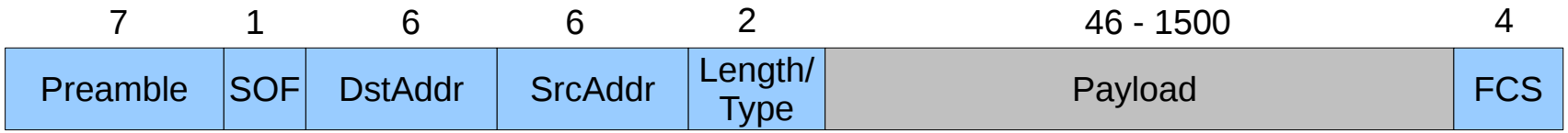
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# Frame format




- The original Ethernet had a slightly different frame format
- Minimum payload size is 46 bytes, smaller messages must be padded (filled up with zeros)
- Maximum payload size is 1500 bytes
- Preamble allows receiver to acquire symbol/bit synchronization
- SOF = Start of Frame
- FCS = Frame Check Sequence, it is a CRC checksum
- Note: there is **no** information as to whether padding happened!

# Ethernet Addresses

- Address fields are 48 bits long
- Each Ethernet adapter has an address, stored in hardware
- Addresses are required to be **globally unique**
  - Each vendor has own address range, assigns addresses from that
  - Nowadays adapters with programmable address available
- Address representation: six colon-separated bytes in hex, e.g.:

  `00:0c:29:10:fb:f3`
- Special addresses and address ranges:
  - Broadcast address: `FF:FF:FF:FF:FF:FF`
  - Addresses with the least significant bit of the leftmost byte set to `1` but different from broadcast address are multicast addresses, e.g., 01:00:5E:12:34:56
  - Addresses with the least significant bit of the leftmost byte set to `0` are unicast addresses, e.g. 02:1A:2B:3C:4D:5E

# The Length/Type Field

| Type field | Protocol |
|---|---|
| 0x0800 | IPv4 |
| 0x0806 | ARP |
| 0x809B | AppleTalk |
| 0x86DD | IPv6 |
| 0x8863 | PPPoE Discovery |
| 0x8864 | PPPoE Session |

- When the L/T field carries value $\geq$ 0x0600 it is interpreted as a type field, indicating the higher-layer protocol that is encapsulated in the Ethernet frame
- The type field therefore provides **protocol multiplexing**
- Table shows some example type values
- When L/T carries a value $\leq$ 1500, it indicates the length of the payload, type field is in the first two bytes of the payload
- When L/T carries value $>$ 1500, frame length is inferred from PHY (e.g. idle gap between frames)

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# Ethernet PHYs

- The Ethernet standard specifies several different physical layers, using different transmission media and adapters:
  - Coaxial cables
  - Twisted pair cables
  - Fiber cables
- The standard also specifies several transmission rates: 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, 40 Gbps, 100 Gbps
- Ethernet supports fundamentally two types of topologies:
  - Broadcast topologies (almost disappeared nowadays)
  - Switched Ethernet

# Broadcast Topologies

- All stations share capacity of the medium, MAC is needed
- Only one successful transmission at a time across the network
- In Ethernet:
  - Bus topology (with coaxial cable)
  - Hub-based topology (e.g. using twisted-pair cable)
- Only used in the 10 Mbps PHYs
- Referred to as half-duplex Ethernet in the following

# Switched Topologies

- Stations are attached to (possibly cascaded) switches via point-to-point links, no MAC needed on those links
  - The MAC is nonetheless “used”, since it is present anyway
- These links have separate cables allowing parallel transmissions in either direction, i.e. stations can transmit and receive at the same time (full-duplex operation)
- Parallel transmissions from different stations possible too, contention occurs for resources in switches (e.g. buffer memory)
- Used for 100 Mbps and higher PHYs
- Referred to as switched Ethernet in the following

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# Half-duplex MAC Protocol

- Half-duplex Ethernet uses CSMA/CD
  - Carrier-Sense-Multiple-Access with Collision-Detection
- Major assumptions:
  - Parallel transmissions lead to collisions
  - Stations monitor the medium for collision while transmitting
    - Ethernet checks the voltage on the medium, a collision is inferred when this voltage exceeds voltage applied by the transmitter alone

# MAC Protocol

1. New frame arrives at the MAC of a station, set `coll` to zero
   - `coll` is a local collision counter, each station has its own
2. Station performs a carrier-sense operation
3. When the medium is idle, transmission starts immediately
4. When medium is busy:
   1. Listen until channel becomes idle again
   2. Start transmitting
5. While transmitting, check for collision
6. If collision is detected:
   1. Abort frame transmission, send a short jamming signal
   2. Increase counter `coll`, counting # of subsequent collisions
   3. If `coll` $>$ 16 then drop frame, set `coll` to zero, stop
   4. Wait for random time (the **backoff time**), then go to step 2
7. If no collision detected, transmit whole frame, set `coll` to zero

# Computation of Backoff Time

- Draw a random integer number uniformly from the interval

$$\left[0, 1, \ldots, 2^{\min\{10, \mathtt{coll}\}} - 1\right]$$

  which is called the **backoff window**
- The actual backoff time is computed by multiplying this integer with the pre-defined **slot time**
- Slot time is a common parameter, just large enough to cover maximum round-trip time and processing delays

**Important Point**

The backoff window size and therefore the average backoff time doubles after each collision, until 10 collisions have been observed! This is called the (truncated) binary exponential backoff algorithm!

**Question**: Motivation for this algorithm?

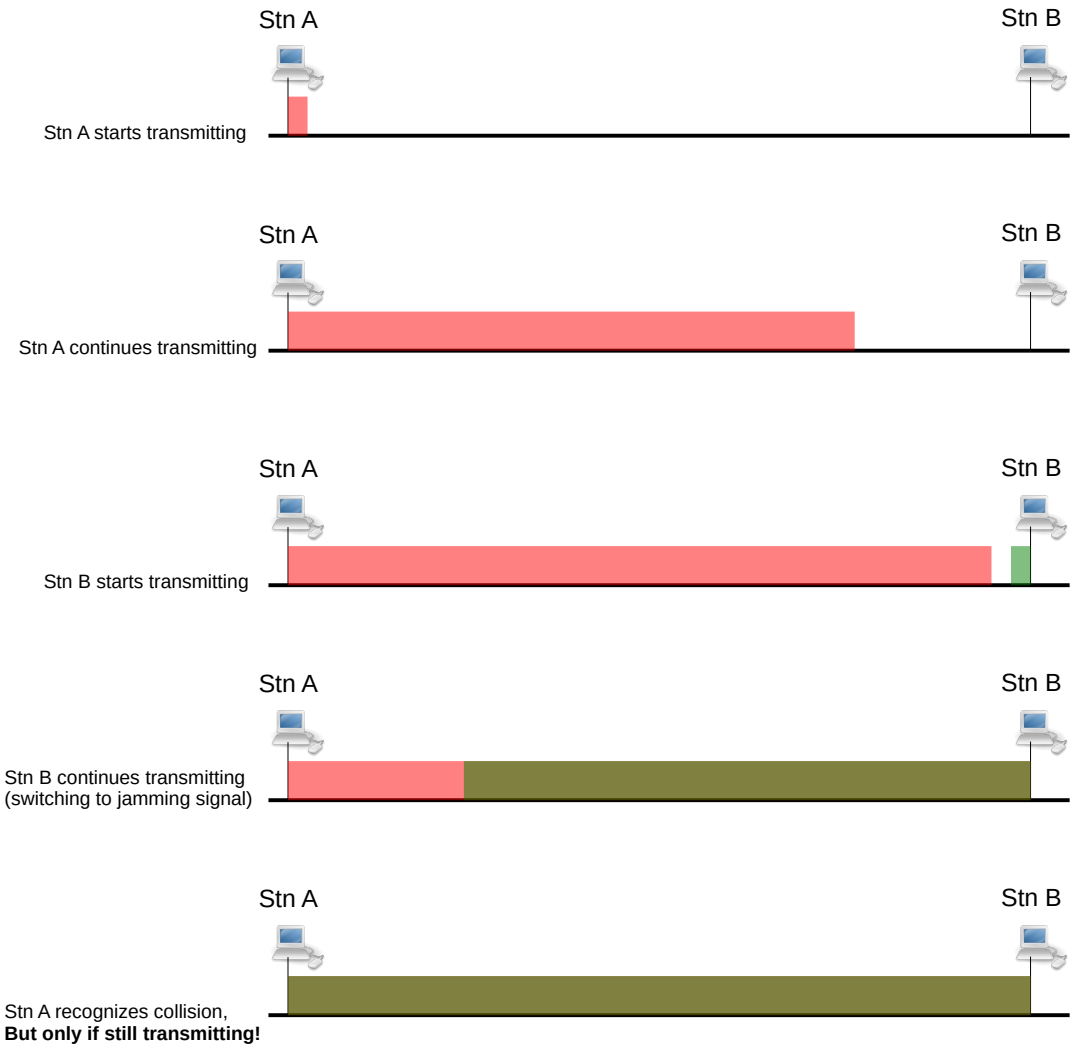
# Minimum Frame Size / Slot Time

- Minimum frame size (64 bytes for 10 Mbps Ethernet) is chosen to ensure that transmitter can indeed detect collision on a cable of given length **while it is still transmitting**
- Rationale: a frame must be long enough to still being transmitted after the maximum round-trip delay plus small processing times introduced by repeaters / hubs
- Requirement for maximum round-trip delay can be understood from worst-case example (next slide)

**Important Point**

Smaller slot times / minimal frame sizes can be achieved by restricting the maximum length of an Ethernet segment!

# Minimum Frame Size (2)



# Half-duplex MAC Protocol – Important properties

- Very short medium access delays under light load
- Under heavy load and for many stations collision rate increases
  - Ethernet should not be operated beyond 50 - 60% load!

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4 Random Access Protocols

5 Ethernet

6 Bridges, Switches

# Coupling LANs

- It is often required to couple existing LANs, e.g. upon merging different departments
- Coupling requires special **coupling devices**
- Types of coupling devices:
  - PHY layer: repeaters, hubs
  - MAC layer: bridges, layer-2 switches
  - Network layer: router
  - Application layer: gateway
- See [24], [35]

# Outline

1 LANs

2 MAC Fundamentals

3 Orthogonal MAC Schemes: FDMA and TDMA

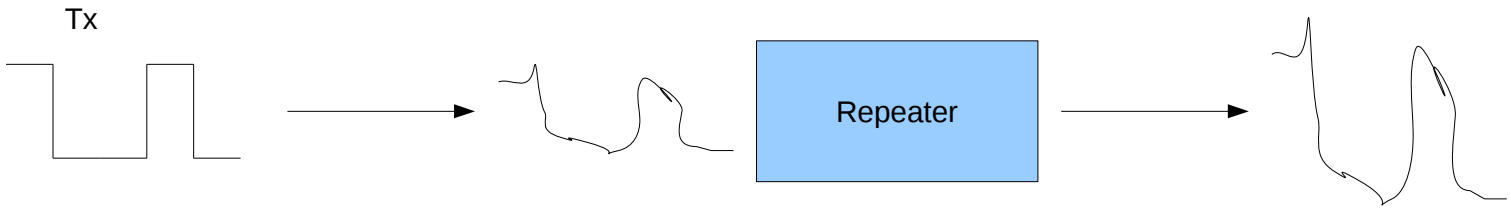
4 Random Access Protocols

5 Ethernet

6 Bridges, Switches
- Repeaters and Hubs
- Bridges and Switches

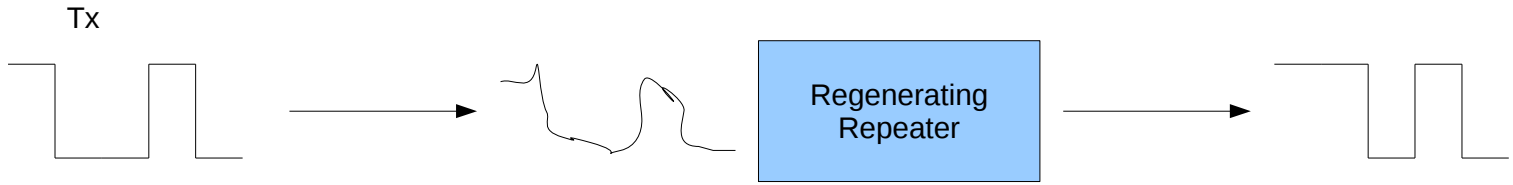
# Repeaters

- A **repeater** amplifies a signal on the analog level
  - Any noise present in the signal is amplified as well
  - Repeaters add their own noise
  - Repeaters are agnostic to any protocol or modulation scheme
  - They can create slight delay (order of $\mu$s and less)

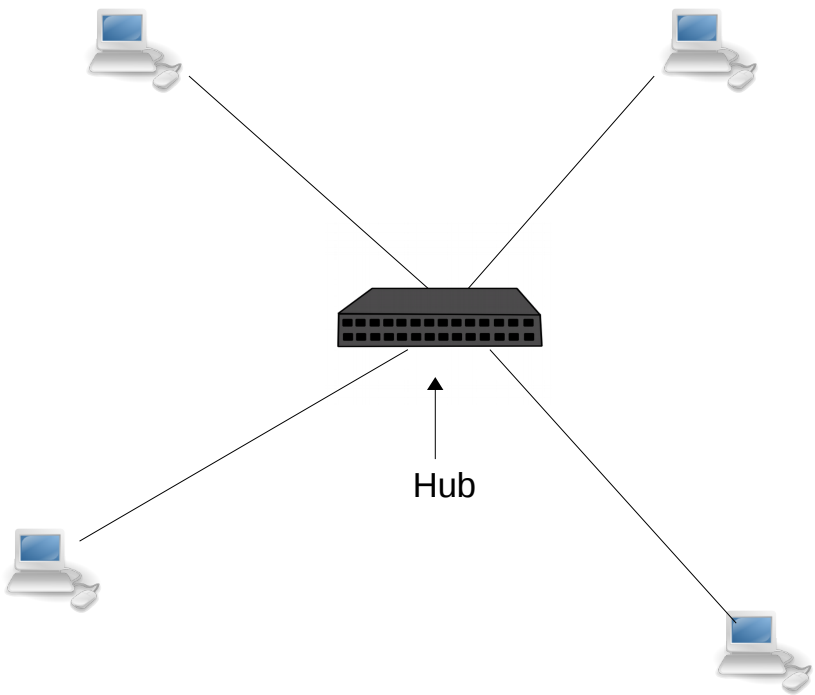


# Regenerating Repeaters

- A **regenerating repeater** demodulates an incoming signal symbol-per-symbol and modulates it again
  - It does not look beyond one symbol at a time
  - No interpretation whatsoever of protocol fields is done
  - Especially, no error checking / error correction is done, regeneration can introduce errors



# Hubs




- Hubs are centralized repeaters, they broadcast signals coming in on one port to all other ports
- No interpretation of the incoming frame is done, none of its fields is evaluated
- All stations are attached with one transmit and one receive line
- A hub creates a broadcast medium on a physical star
- Hubs can be cascaded
- It may be regenerative or not
- **Question**: Advantage over bus?

# Outline

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- Repeaters and Hubs
- Bridges and Switches

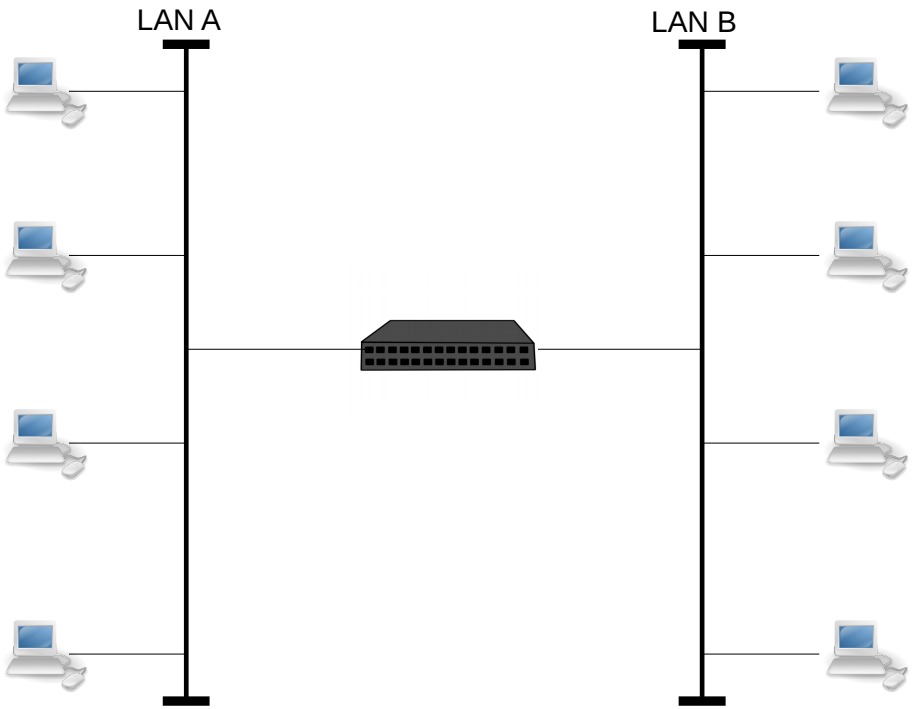
# Bridges

- Bridges interconnect LANs on the MAC layer
- They mostly connect LANs of the same type (e.g. Ethernet), but bridges connecting LANs of different types **but the same MAC address structure** (e.g. Ethernet – Wireless LAN) also exist
- We focus on bridges interconnecting (broadcast) Ethernets
  - Important: Ethernet frames carry an Ethernet source address and an Ethernet destination address in their frame header
- A bridge can connect several LANs
- Bridges can be cascaded

**Important Point**

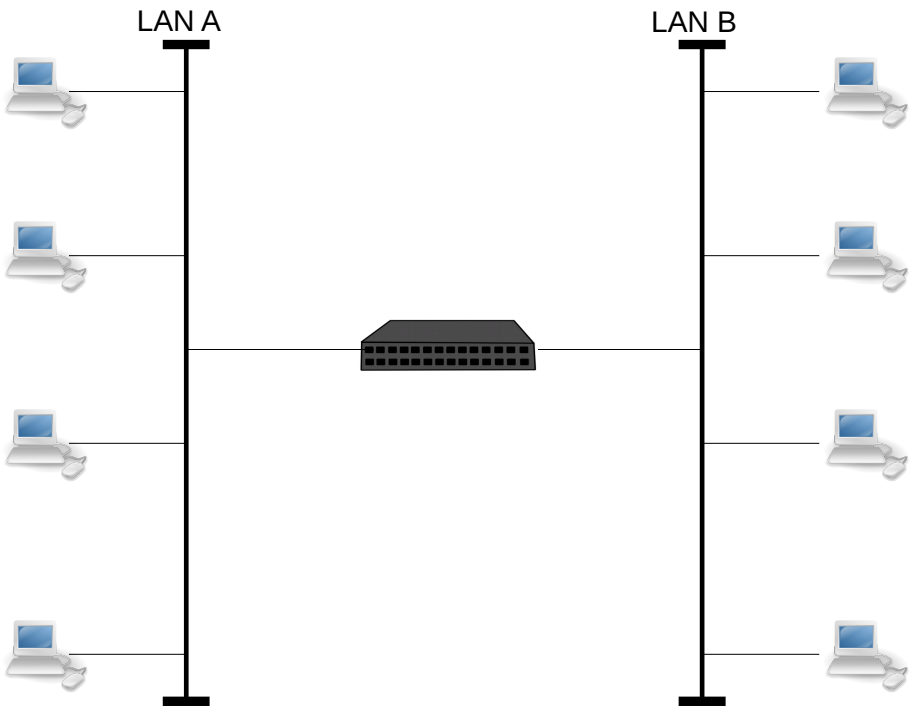
Bridges understand and interpret fields related to the MAC protocol (e.g. address fields), repeaters / hubs do not!

# Basic Operation




- When bridge receives frame from LAN A, it checks for correctness, buffers it and checks MAC destination address (`dst`)
- `dst` on LAN A: bridge does nothing
- `dst` on LAN B or `dst` unknown: bridge transmits frame on LAN B, **following the rules of the MAC protocol!**
- Same in direction B $\rightarrow$ A
- Bridge does not modify any frame, nor does it encapsulate them
- Stations need not be aware of the presence of bridges (**transparency**)

# Basic Operation (2)

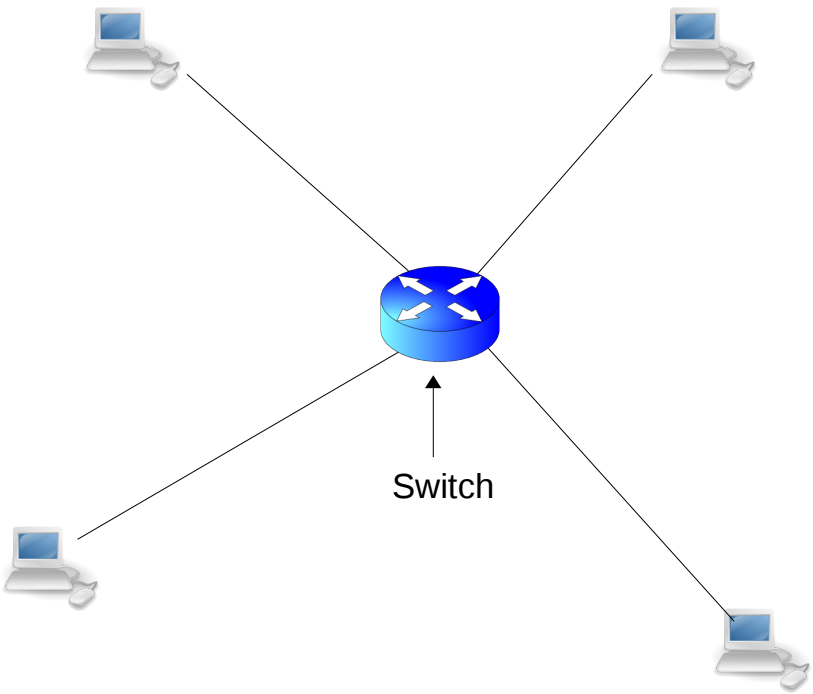



- How does the bridge know which station is in which LAN?
- From reading a frames source address field (`src`) the bridge can learn on which bridge interface a source can be reached
- When a bridge receives a frame with `dst` not having been observed so far, it unconditionally re-transmits the frame on all interfaces except incoming one
- That is dangerous when several bridges are used and loops are present
- **Q**: Can you give an example?
- Additional mechanisms are needed to deal with loops – the Rapid Spanning Tree Protocol (RSTP, IEEE 802.1w) closes ports on bridges to remove loops

# Some Reasons to Use Bridges

- **Reliability**: By keeping LANs separated and only interconnected by a bridge, failures in one LAN do not affect others
- **Performance**: By carefully evaluating addresses, bridges can confine traffic local to one LAN to that very LAN, enabling parallel local transmissions in different LANs
  - Repeaters / hubs cannot achieve this “traffic separation”!
- **Security**: Similarly, traffic local to one LAN cannot be eavesdropped in the other LAN

# Layer-2 Switches




- A switch is a centralized element, forwarding frames **only to the correct output port**
- Stations are attached to switch via point-to-point links with separate transmit/receive lines (full-duplex)
- No broadcast medium anymore!
- Switches can process frames to distinct destinations **in parallel**, switches can increase network capacity as compared to LANs with broadcast medium
- Frames arriving in parallel for the same destination are buffered (output buffering)
- Switches are transparent to stations
- Nowadays almost all Ethernet installations use switches

# Layer-2 Switches (2)

- Difference to a hub:
  - A hub builds a broadcast medium, only one station can transmit at a time without collisions
  - A switch can accept up to $N$ parallel transmissions, where $N$ is the number of stations in the LAN
  - Each station can send/receive packets at full medium capacity

**Important Point**

Switches remove the “shared medium” assumption and the need for a MAC, but now the stations contend for the resources of the switch (switching capacity, buffer memory)!

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